

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2023
 Programme: BE Full Marks: 100
 Course: Engineering Economics Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Why we need to study engineering economics? State and explain law of demand and supply. 2+5
- b) What do you mean by IRR? Evaluate IRR of the following project and decide whether the project acceptable or not? Where MARR 8% per year. 8
- Initial investment =Rs. 5,000,000
 Expected life =10 Years
 Salvage value =Rs. 30,000
 Annual revenue =Rs. 120,000
- a) How do you measure following cost concept in your engineering project; direct cost, indirect cost, average cost, variable cost, marginal cost, sunk cost, opportunity cost? 8
- b) From the following project information select the best project using capitalized worth method. MARR = 10% 7
- | Items | X | Y | Z |
|--------------------|--------|--------|--------|
| Initial Investment | 50,000 | 40,000 | 30,000 |
| Annual Revenue | 20,000 | 15,000 | 14,000 |
| Annual Expenses | 15,000 | 10,000 | 8,000 |
| Useful life year | 5 | 7 | 9 |
| Salvage value | 1000 | 500 | 0 |
- a) What is the future equivalent of Rs. 500,000.00 per year uniform cash flows for 10 years if nominal rate is 10% compounded continuously? 8
- b) Discuss economic theory and ecological footprint. In the pursuit of achieving global sustainable goals, what vital role should engineers play to overcome ecological limits and promote environmentally responsible practices? 7
- a) Find the B/C ratio by both methods. Use AW method. Investment = Rs.2,50,000; Annual benefits = Rs.75,000; Annual cost = Rs.15,000; salvage value = Rs.25,000; MARR = 12%; Number of years = 10. 8

- b) What do you mean by project financing and project funding? Explain with some example. 7

- a) By using PW model & repeatability assumption select the best project. 7

Project	A	B
Initial Investment Rs	4,00,000	6,00,000
Annual Revenue Rs.	30,000	35,000
Annual O & M Rs.	3,000	4,000
Useful life in year	6	8
Salvage value Rs	4,000	7,000
MARR	12%	12%

- b) What are financial statements? Explain the types of financial statements with their formats. 8

- a) Perform sensitivity analysis of the following project over the range of $\pm 20\%$ for 8

- i. initial investment
 ii. annual revenue
 iii. useful life &
 iv. MARR.
 Initial investment = 11,500
 Annual revenue = 3,000
 Salvage value = 1,000
 Useful life = 6 years
 MARR = 12%

- b) A new electric sawmill for cutting small pieces of lumber in a furniture manufacturing plant has a cost basis of Rs. 5,000 and 8 year depreciable life. The estimated salvage value of the saw is zero at the end of 4th year cumulative depreciation charge of 7th year and book value of 6th year using double declining balance method. 7

Write short notes on: (Any two)

- a) Importance of financial statement and accounting equation.
 b) Financial Ratios
 c) Ecology and economic development.

2x5

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2023
 Programme: BE Full Marks: 100
 Course: Engineering Economics Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Discuss Engineering Economics and describe Principles and applications of Engineering Economics. 7
- b) Define manufacturing, non-manufacturing, opportunity and sunk cost with suitable examples. 8
2. a) Mr. Hari Lamsal needs Rs2500000 at the end of 15th year in his bank account. How much money does he need to deposit at the end of every year for 10 years to meet his requirement if his bank provides him 12% interest compounding annually? 8
- b) Evaluate the following project under IRR method and also show unrecovered investment balance in tabular and graphical forms. 7

Initial investment: Rs 1500000
 Annual revenue: Rs450000
 Annual cost: Rs50000
 Life: 6 years
 Salvage value: 60000
 MARR: 13%
3. a) Calculate both types of BC ratios from the following information and also give your decision whether the project should be selected or not. 7

Initial investment: Rs 800000
 Annual revenue: Rs 150000
 Operating and maintenance cost: Rs25000
 Life: 10 years
 Salvage value: 20000
 MARR: 13%
- b) What do you understand by payback period? Calculate simple and discounted payback period from the following information. 8

Initial investment 1000000 MARR 10%

Year	1	2	3	4	5	6
Net Annual Revenue	250000	345000	320000	280000	125000	210000

4. a) Project A and B are mutually exclusive projects. By using repeatability method evaluate the both projects. 8

Items	Project A	Project B
Initial investment	Rs200000	Rs 300000
Annual Revenue	Rs 75000	Rs 85000
Annual Cost	Rs 10000	Rs 12000
Useful life	6 years	8 years
Salvage Value	Rs 50000	Rs 80000

MARR 9% for both projects

- b) Discuss Financial Ratio and explain the followings in detail. 7
 - i. Current Ratio
 - ii. Debt Ratio
 - iii. Total asset Turnover ratio
 - iv. Inventory Turnover ratio
5. a) Explain causes of depreciation. A machine costing of Rs. 10,000, estimated Salvage Value of Rs. 500 at the end of 5 year and MARR is 12%. Find depreciation each year by using: 8
 - i. Sinking Fund Factor method.
 - ii. SOYD method.
- b) Pathivara Engineering Works Company has following capital structure. 7

Common Stock Rs 1000000
 Bond Rs1500000
 Preferred Stock Rs 500000
 Total capital Rs 3000000

The company's cost of common stock is 21%
 Cost of bond before tax is 18% and tax rate is 40%
 Cost of preferred stock is 17%

Calculate weighted average cost of capital (WACC) of the company.

6. a) Perform Sensitivity analysis of the following Project over the range of $\pm 30\%$ in i) Initial investment ii) Annual revenue iii) Useful life and iv) Salvage Value: 8

Initial Investment (Rs)	14,500
Annual Revenue (Rs)	4,000
Salvage Value (Rs)	10,000
Useful life (Years)	6
MARR	20%

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Attempt all the questions.

1. a) Describe engineering economics. Explain the different principles of engineering economic. 5+2
- b) You make 10 annual deposits of 1,000 each into a bank account paying 5% interest per year. The first deposit will be made one year from today. How much money can be withdrawn from this account after 10 years? What is the present worth of that sum? 8
2. a) Discuss any four basic cost concepts used in engineering economics. Explain why these cost concepts are important in engineering economic decisions. 7
- b) A piece of new equipment has been proposed by engineers to increase the productivity of a certain manual welding operation. The investment cost of equipment is Rs 1,00,000 and has market value of Rs. 20,000 at the end of five years. The equipment will generate Rs. 32,000 per year after operation costs. The company is willing to accept the project if it will earn at least 20% per year. Evaluate IRR of this equipment and suggest whether the equipment is feasible or not? 8
3. a) Find both types of BC ratios from the following project and also suggest if it is feasible or not. 7
Initial investment Rs 1500000
Annual revenue Rs 300000
Annual cost Rs 60000
Estimated useful life 8 years.
Salvage value Rs 120000
MARR 12%
- b) A mobile company is taking quotations for purchase, installation and operation of microwave towers for long period. If MARR is 15%, determine the best alternative project using repeatability assumption and PW method. 8

Quotations	Project A	Project B
Equipment cost (Rs)	6500000	5800000
Installation (Rs)	150000	200000
Annual maintenance (Rs)	100000	125000
Annual extra charge (Rs)	0	50000
Salvage value (Rs)	0	0
Life (yrs)	15	10

4. a) Select the best project by using BCR method using FW formulation when MARR is 8.5%. Use incremental analysis if necessary.

Year	Project A	Project B
Initial Investment	6,00,000	10,00,000
Annual Revenue	3,00,000	3,50,000
Life	6	6
Salvage Value	1,20,000	1,80,000

b) What is break even Analysis? Phlewa cement shop open the market to sell 10,000 bags of cement in one month, the shop purchase the cement at the rate Rs 650 in bulk and sell at the rate of Rs 800. What will be the break-even quantity, if the shop sold all the quantities what will be the profit or loss, when the shop pay Rs 15,000 for rent and Rs 5000 for salary? Initial Investment Was Rs 6500,000.

5. a) ABC Company is planning to purchase a machine costing of Rs 500000 having estimated salvage value of Rs 45000 and useful life 6 years. Calculate every year depreciation amount and book value under Sinking Fund and SOYD depreciation method, if MARR is 12%.

b) Discuss economic theory and ecological footprint. What role an engineer should play for overcoming ecological limit and achieve global sustainable goals?

6. a) What is financial ratio? Explain major financial ratios that can be applied in decision making process in business.

b) What are the funding mechanism of Nepal government to the project while talking about project funding mechanism?

Write you agree existing mechanism or need to improve it?

7. Write short notes on: (Any two)

 - Balance sheet
 - Ecological limit and sustainable development
 - Project risks

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
 Programme: BE Full Marks: 100
 Course: Mobile and Wireless Communication Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is DECT? Explain the significant changes in the evolution from 1G to 4G mobile communication systems. 8
- b) Why hexagonal geometry are used in the design of cellular system. Prove that the co-channel cell reuse ratio is given by $Q = \sqrt{3N}$. 7
2. a) What is handoff and its types? Explain different types of channel assignment strategies used in cellular system and their advantages and disadvantages. 8
- b) Consider a transmitter which radiates a sinusoidal carrier frequency of 1850 MHz. For a vehicle moving 60 mph, compute the received carrier frequency if the mobile is moving
 - (i) directly towards the transmitter,
 - (ii) directly away from the transmitter,
 - (iii) in a direction which is perpendicular to the direction of arrival of the transmitted signal.
 7
3. a) Explain Friis Free Space Loss model and define free space path loss. A Transmitter produces 80 W of power, express the power in units of
 - i) dBm ii) dBW
 7
- b) Why is speech coding needed? Explain the basic concepts of VOCODERS. 8
4. a) Write the waveforms to explain MSK transmission for a binary sequence 1010101. 7
- b) Why equalization is need in wireless communication? Explain different diversity techniques used in wireless communication. 8

5. a) What is Multiple Access Technique? Explain Space Division Multiple Access (SDMA). 7
- b) Explain forward CDMA channel in details. Write the major function of each forward channel. 8
6. a) The most remarkable feature in GSM is on-the-air privacy, justify how it is possible? Also, explain the databases used in GSM architecture. 8
- b) For a given (6,3) linear block code the generator matrix G is given as

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$
 - i. All corresponding code vectors
 - ii. Minimum Hamming Distance
 - iii. Verify that code is a single error correcting code
 - iv. Parity check matrix
 - v. Determine the transmitted codeword if received codeword is 100011
 7
7. Write short notes on: (Any two) 2×5
 - a) WiFi Vs. WLAN
 - b) Link budget design
 - c) Hybrid Spread Spectrum Multiple Access Technique

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 Course: Mobile and Wireless Communication Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain in brief about the evolution of wireless technologies 1G to 5G in mobile communication. 8
- b) Why does minimizing reuse distance maximize spectral efficiency of a cellular system? For a seven-cell reuse pattern, find the minimum distance between centres of co-channels cells. Area of each cell is uniform and is equal to 23 square km. 7
2. a) Define handoff margin with appropriate figure. A telephone network company needs to expand its capacity based on demand on a city. A group of engineers was selected to find the solution. Among the solution sectoring and cell splitting were major technique for expansion purpose. Being a cellular planning engineer, which option do you think is best and why? 7
- b) Evaluate the feasibility of a 5 km link, with one access point and one client radio. The access point is connected to an antenna with 10dBi gain, with a transmitting power of 20 dBm and a receive sensitivity of -89dBm. The client is connected to an antenna with 14dBi gain, with a transmitting power of 15dBm and a receive sensitivity of -82dBm. The cables in both systems are short, with a loss of 2dB at each side at the 2.4 GHz frequency of operation. 8
- a) Explain in brief about free space propagation model with necessary derivations and expressions. 8
- b) Explain in brief about the three basic propagation mechanism in a mobile communication system. 7
- a) Why modulation technique is required in communication system? Explain OFDM. 7
- b) What is Space Division Multiple Access (SDMA)? Explain any two hybrid spread spectrum multiple access technique which minimizes the effect of near far effect. 8

- a) Describe a signal processing operation that minimizes the effects of ISI. Explain various diversity techniques. 8
- b) What do you mean by multiple access? Explain any three multiple access techniques in wireless communication. 7
- a) Explain with necessary diagram system architecture of GSM. Mention functions of various blocks in the architecture. 8
- b) Explain in detail the design principles of LTE. 7

Write short notes on: (Any two)

2×5

- a) Android
- b) Handoff
- c) Advantages of Wireless Communication

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Mobile and Wireless Communication

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare and contrast the first, second, third and fourth generation of mobile communication standards in terms of technology advancement. 7
- b) Why hexagonal cells are used for planning of cellular system. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + j^2 + ij$ is the cluster size. 8
2. a) Explain the handoff process and its types in cellular system. 7
- b) Determine the propagation path loss for a radio signal 900 MHz cellular system operating in a large urban city, with a base station transmitter antenna height of 100m and mobile receiver antenna height of 2m. The mobile unit is located at a distance of 4 km. Use Hata propagation path loss model. 8
3. a) Explain basic radio propagation mechanisms. What is the Doppler spread in small scale fading? 7
- b) Find the linear block encoder G if code generator polynomial $g(x) = 1 + x + x^3$ for a (7,4) code. 8
4. a) Explain the transmission and detection process of QPSK modulation technique. 7
- b) Describe vocoders with block diagram? Briefly explain different kind of vocoders. 8
5. a) Why Equalization and Diversity is need in Wireless Communication? What are the advantages of equalization technique? 8
- b) What is Space Division Multiple Access (SDMA)? Explain any two hybrid spread spectrum multiple access technique which minimizes the effect of near far effect. 7
6. a) Explain with necessary diagram system architecture of GSM. Mention functions of various blocks in the architecture. 8

- b) Explain Space Division Multiple Access (SDMA) with its benefits in advance wireless communication system. 7

7. Write short notes on: (Any two) 2×5

- a) Trunking and Grade of Service (GOS)
- b) Rake Receiver
- c) Frequency Division Multiple Access (FDMA)

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2024
Programme: BE		Full Marks : 100
Course: Mobile and Wireless Communication		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the features introduced from second to fifth generations of cellular communication system. 8
- b) Why hexagonal cells are used for planning of cellular system? Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + j^2 + ij$ is the cluster size. 7
2. a) Explain handover process in cellular system. Write about various practical handover consideration. 8
- b) State the difference between large scale and small scale propagation model. Explain the different propagation mechanisms which have impact on propagation in mobile environment. 7
3. a) Calculate the mean path loss using Okumura's model for $d = 50 \text{ km}$, $h_{te} = 100 \text{ m}$, $h_{re} = 10 \text{ m}$ in a suburban environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz , find EIRP (dBm) and the power at the receiver where gain at receiving antenna is 10 dB . 8

OR

Consider a transmitter which radiates a sinusoidal carrier frequency of 1850 MHz . For a vehicle moving 60 mph , compute the received carrier frequency if the mobile is moving

- i. directly towards the transmitter,
- ii. directly away from the transmitter,
- iii. in a direction which is perpendicular to the direction of arrival of the transmitted signal.
- b) Assume a receiver is located 10 km from a 50 W transmitter. The carrier frequency is 900 MHz , free space propagation is assumed, $G_t = 1$, and $G_r = 2$, find: 7
 - i. the power at the receiver,

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- ii. the magnitude of the E-field at the receiver antenna
- iii. the rms voltage applied to the receiver input assuming that the receiver antenna has a purely real impedance of 50Ω and is matched to the receiver.
4. a) The generator polynomials of a $1/3$ binary convolution encoder is given by $g_1(x) = x^3 + x^2 + 1$, $g_2(x) = x^3 + x$, $g_3(x) = x^3 + 1$. Calculate the encoded bit sequence for input sequence 01101 . 8
- b) Why we need speech coding techniques? Explain the basic concept of VOCODER. 7
5. a) What is the advanced feature of non-linear equalizer over linear equalizer? Explain the blocks of one non-linear equalization technique. 8
- b) What is Multiple Access Technique? Explain Frequency Hopped Multiple Access (FHMA). 7
6. a) Explain with necessary diagram system architecture of GSM. Mention functions of various Blocks in the architecture. 8
- b) Explain forward CDMA channel in details. Write the major function of each forward channel. 7
7. Write short notes on: (**Any two**) 2×5
 - a) Frequency Re-use
 - b) OFDM
 - c) Wimax

Page 2 of 2

- b) Prepare Balance sheet of Himalayan Tea Company as on 31st December 2023 from the following transaction.

Common Stock	1000000	Land and Building	1362000
Inventories	45000	Cash	75000
Vehicles	25000	Creditors	65000
Bank overdraft	200000	Debtors	120000
Prepaid insurance	25000	Outstanding salary	35000
Marketable securities	28000	Bond and Debenture	350000
Provision for taxation	55000	General Reserves	180000
Furniture	85000	Machinery	120000

7. Write short notes on: (Any two)

- Nominal and Effective interest rate
- VAT
- Ecological limit and sustainable development

Economics-2023 Fall

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year: 2021
 Programme: BE Full Marks: 100
 Course: Social and Professional Issues in IT Pass Marks: 45
 Time: 3hrs.

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Attempt all the questions.

- Discuss the contribution of Conrad Zuse in the field of computing. 8
 - Discuss the relationship between society and the technology with real world examples. 7
- Is "Gender Divide" making the Digital Divide even larger? Explain the cross-gender communication in the Internet. 8
 - What are the common Legal and Moral issues faced by a Software Engineers. Explain with example. 7
- Why is Engineering termed as a profession? Give reasons. 8
 - What do you mean by a professional code of ethics? Explain the code of ethics of the Nepal Engineering Council. 7
- "Software risks are considered critical in case of life critical systems as the developer is the only one to blame if an accident occurs." Do you agree or disagree? Justify your opinion. 8
 - Is whistleblowing a professional act? What are the conditions required to blow a whistle? 7
- What do you mean by Intellectual Property Rights. Explain different types of Intellectual Property in detail. 8
 - Cyber Terrorism has been increasing with the advancement of technology. Express your view on it with the measures that can be taken to control this issue. 7
- What do you mean by Cyber crime and Digital Forgery? Discuss your view on current situation of online harassment and measures to prevent it. 5

- b) Read the following case study and answer the questions at the end. 5×4
= 20

Virtual Class

"... In 1978 ..." "Like most people with real jobs, I had to travel a lot which meant that I was frequently out of town on the nights I was supposed to be teaching. So I had to develop some strategies for coping with this problem. One was to participate in the class via an audio conference. I would have someone set up a speakerphone in my classroom and call in from wherever I happened to be. I could give lectures and engage students in discussions. It worked quite well." "Another strategy was to ask colleagues to "cover" for me by going to the class and giving a guest lecture. I would pick out people with expertise on the topic to be taught so their contribution to the class was usually very worthwhile and enjoyed by the students. I found that having 5 or 6 guest lectures in a course made it much more interesting to the students than being taught by a single person." "The third strategy involved the use of computer bulletin board systems to provide a way for students to contact me and each other via email and online conferences. Not only was this useful for me, but it was helpful to the students as well since many of them also traveled a lot and this gave them a way to keep up to date with course work."

Questions:

- i) In what aspects do you consider this class as a virtual one?
- ii) Can you think of applying the online education in this context? How?
- iii) Analyze the implications of technology on students and professor in this case.
- iv) If you were the professor, what improvements would you make in this situation?

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2021
Programme: BE		Full Marks: 100
Course: Social and Professional Issues in IT		Pass Marks: 45
		Time : 3hrs.

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Attempt all the questions.

1. a) Discuss some of the major breakthroughs in the history of hardware, software and networking. 8
 b) Define E governance. Discuss various e governance initiatives taken by government of Nepal and its various challenges. 7
2. a) What are the social implications of Information Network? Explain in details. 7
 b) What do you mean by a professional code of ethics? Explain the code of ethics of the Nepal Engineering Council. 8
3. a) How risks in critical systems can be minimized? Discuss with examples. 8
 b) "Accuracy vs Democracy" has been a raising concern in the internet. Express your views with your preference. 7
4. a) What do you mean by Privacy risk? Illustrate the importance of censorship in cyberspace with relevant examples. 8
 b) What is Ethical Relativism? Compare and contrast "Utilitarianism and Deontological" theory in the light of ethical relativism. Give examples to support your answers. 7
5. a) What do you mean by Digital Forensic? Explain about Digital Forgery and Phishing. 8
 b) What are the different social-media related cyber-crimes that are commonly seen in Nepal? How can the government handle this issue? 7
6. Read the following case study and answer the questions at the end.
 True Soft is a large company that sells computers, computer components, and software. Rajesh is hired as an entry-level software engineer at True Soft. His first project was to assist in writing the code for True Soft's new hard disc controller. He had previously worked on a similar system interning

at a start-up and had written a code which greatly enhanced the performance of their product. Rajesh quietly re-uses this same code in the True Soft product, and does not think to tell anyone that he has used the code from his last job. His manager is thrilled with the speed improvements this code brings to the product.

Before the product is released, it has to undergo a four-month long quality assurance process review. During the review of the product, it was found the code which Rajesh developed had been copyrighted by the start-up he had previously worked for. Even though Rajesh had developed the code, his previous company still owned the intellectual property rights to it.

When his manager informed Rajesh of the problem, Rajesh admits he did not realize he had made a mistake because he was not familiar with copyright laws. Rajesh then goes on to explain that the start-up he used to work for is now out of business and is unsure if True Soft would be able to get in contact with the owner of the copyright. If True Soft can't use Rajesh's code, then it will have to rewrite the entire code of the product, delaying its release by many months.

- a) Discuss prevalent laws in such situations in context of Nepal. 8
 - b) If you were in position of Rajesh, what would you do in such situation? 7
- Explain your answer with ethical lenses of rights, utilitarianism and deontology?

7. Write short notes on: (Any two) 2x5

- a) Digital divide
- b) Email Privacy
- c) Descriptive Claims and Normative Claims

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Attempt all the questions.

1. a) Define Computing. Explain the major milestones in the history of Computing? 8
- b) Advancement of Technology is today's demand of society. Briefly explain the impact of technology on the society? 7
2. a) What is Digital? Is it actually not a thinking divide? Give your opinion. 7
- b) Define Philosophical and professional ethics. Explain the descriptive & normative claims. 8
3. Chinese web users are being denied access to a range to a range of religious sites based abroad as the Chinese government's "Golden Shield" firewall is used to censor The Internet. While blocking undesirable sites promoting pornography and violence, the firewall also limits religious web content such as sites related to the Dalai Lama, the Falun Gong Cult and various Buddhist and Muslim movement. Also blocked are sites covering persecution of religious communities in China and number of Catholic sites, including the website of the Hong Kong diocese and the Divine Word missionaries in Taiwan. Not blocked are sites in European languages covering religious freedom issues, even those covering repression within China. While overall Internet usage in China may be low by developing country standards, it has been rapidly growing, especially in the capital of Beijing and the coastal region. The official China Internet Network Information Center puts the number of Chinese with access to the Internet in June 2003 at 79.5 million. 5+5+5

The number has been doubling every six months. Internet censorship is part of a comprehensive attempt to censor all means of communication. While printed publications have long been censored in China.

authorities also have tried to keep up with technological developments. The Global Internet Policy Initiative warned in June of new technology from a Chinese firm that monitors "subversive" text messages sent by mobile phone.

Questions:

- i. The government practices the Right to Information Act and the government itself denies access to a range of sites. Isn't it conflicting to one another?
 - ii. Based on the above scenario provided in the case above, do you think it is correct to enforce censorship in the cyberspace? Elaborate your opinion.
 - iii. Put your views on the relevancy of the control over the internet?
- a) What are Professional Rights and Responsibilities? How can you avoid the whistle blowing in the organization? 8
 - b) Put your views on accuracy vs democracy in Internet with relevant 7
 - a) What is personal privacy? Explain with suitable example how technology is affecting one's personal privacy. 8
 - b) What do you mean by denial of service? Explain different types of malicious program? 7
 - a) What do you mean by digital forensics? Explain different cybercrimes and the way to overcome. Put your own views? 8
 - b) What are the trademarks and trade secrets? How important are they in context of protecting intellectual properties? 7
- Write short notes on: (Any two) 2x5
- a) E- Governance
 - b) Code of Ethics of Nepal Engineering Council
 - c) I.T Policy of Nepal.

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Attempt all the questions.

1. a) How is mathematics related to computing? Explain the relation from the context of History. 7
- b) How has Internet changed the society? Explain. 8
2. **Case Study: Cybercrime-related cases see an alarming rise**
Published at : April 16, 2023
 Nepal's Cyber Bureau says the country has seen a steep rise in cybercrime cases over the past year. Inadequate technical human resource and lack of relevant laws have compounded the problem, the bureau says.
 Over the past eight months, the bureau has registered a total of 4,937 cases, a number higher than all cases registered in the whole of last fiscal year (4486 cases), according to Pashupati Kumar Ray, the bureau's spokesperson.
 The country faced its biggest cyberattack at the end of January this year, which resulted in disruptions of hundreds of government websites across the country. It even halted international travel due to the shutdown of the immigration server.
 At the time, around 1,500 government websites were shut down due to attacks on the central data bank at the Government Integrated Data Centre, according to Ramesh Pokharel, assistant director at the National Information Technology Centre under the Ministry of Communication and Information Technology.
 Meanwhile, the nature of cybercrimes is fast evolving with the advent of new technologies. People have shifted to live online gambling through the Chinese app TikTok and recently, the app has been widely used to lure clients for sex massage.
 This year, financial frauds committed online topped the list, and cases of 'revenge porn' came second. A total of 955 cases of financial fraud were lodged with the bureau, or 20 percent of the total online crimes. The financial crimes include phishing (attempting to acquire sensitive data such